PHYSICS
VI
IIT FOUNDATION

CONTENTS

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MEASUREMENTS

1. Physical Quantity:

The quantities which are measurable are called physical quantities.

Eg : Length, mass, time, speed, etc.

2. Unit

Unit is a standard. To measure any physical quantity one standard should be required i.e., called unit.

Eg : Meter is the standard by which we can measure length.

$\therefore$ unit of length is metre

Eg : Second is the standard by which we can measure time.

$\therefore$ unit of time is second

3. Numerical Value

The number of times a standard quantity is present in a given physical quantity is called Numerical value of physical quantity.

Physical quantity = Numerical value × Unit

Eg : Let length of table = 3 metre

Here 3 is the Numerical value and metre is the standard unit.

4. Systems of Units

S.NO	Name of the measuring system	Basic units		
		Length	Mass	Time
1	F.P.S	Foot	Pound	Second
2	C.G.S	Centimetre	Gram	Second
3	M.K.S	Metre	Kilogram	Second

The fundamental unit of length, mass and time taken together form a system of units.There are four system of units.

(i) British or F.P.S. system. (ii) French or C.G.S system.

(iii) M.K.S system or metric system. (iv) International system or S.I.

In F.P.S. system

The unit of length is foot.

The unit of mass is pound.

The unit of time is second.

In C.G.S system

The unit of length is centimetre.

The unit of mass is gram.

The unit of time is second.

In M.K.S system

The unit of length is metre.

The unit of mass is kilogram.

The unit of time is second.

S.I. system has seven basic units and two supplementary units.

5. **Fundamental quantities :** A physical quantities which are independent of other physical quantity.

   Eg : Length, mass and time

6. **Fundamental units :** Fundamental units are the units for measuring fundamental quantities. These are independent of other units

   Eg: Unit of length - metre

   Unit of mass - kilogram

   Unit of time - second

   These are also called basic units.

7. **Derived quantities :** A physical quantities which are depends on fundamental quantities.

   Eg : Area, volume and density etc.

8. **Derived units :**Derived units are the units of derived physical quantities which are expressed in terms of fundamental units.

   Eg: The unit of speed is expressed by dividing the unit of distance by unit of time.

   Hence speed is a derived quantity and the unit of speed = ms^{-1}

9. Table of multiplication factor, prefix and symbol

S.No	Multiplication factor	Prefix	Symbol
1.	*10	deca	d
2.	*10^2	Hecta	h
3.	10^3	Kilo	K
4.	10^6	Mega	M
5.	10^9	Giga	G
6.	10^{12}	Tera	T
7.	10^{15}	Peta	P
8.	10^{18}	Exa	E
9.	10^{21}	Zetta	Z
10.	10^{24}	Yotta	Y
11.	*10^{-1}	Deci	d
12.	*10^{-2}	Centi	c
13.	10^{-3}	Milli	m
14.	10^{-6}	Micro	μ
15.	10^{-9}	Nano	n
16.	10^{-12}	Pico	p
17.	10^{-15}	Femto	f
18.	10^{-18}	Atto	a
19.	10^{-21}	Zepto	z
20.	10^{-24}	Yetto	y

Rules for writing the symbols of units :

1. The symbol for a unit which is not named in the honour of some scientist is written in lower letter.

   Ex: The symbol for metre is 'm' for kilogram is 'kg' and for second as 's'.

2. The symbol for a unit which is named in the honour of some scientist is written with initial capital letter.

   Ex: The symbol for unit of force (newton) is N.

   The symbol for unit of temperature (Celsius) is °c.

   The symbol for unit of work (joule) is J.

The symbol for unit of power (watt) is W.

3. Full name of the unit named in the honour of scientist is written with lower initial letter.

   Ex: The full name for the unit of force is newton and not Newton.

   The full name for the unit of power is watt and not Watt.

4. A compound unit obtained by multiplying two or more units is written after putting a dot between the individual units or leaving space between the individual units.

   Ex: Unit of momentum is written as N.s or N–s.

5. Negative powers are used for compound units obtained by dividing one unit with another unit.

   Ex: The unit of velocity is m/s. It is expressed as ms^{-1}.

6. A unit in short form is never written in plural.

   Ex: 30 kilogram in short form is written as 30kg and not 30kgs.

1. The gap between two points is measured by length. In S.I. system, the length is measured in unit metre (m).

2. **Multiples of metre**

i) decametre (dam) = 10^1 m = 10 m

ii) hectometre (hm) = 10^2 m = 100 m

iii) kilometre (km) : One kilometre is the one thousand multiple of a metre.

1 km = 10^3 m = 1000 m

3. **Practical units of length :**

i) Astronomical unit (A.U) : It is the mean distance of the earth from the sun.

1 astronomical unit = 1.496×10^{11} m

ii) Light year (ly) : The distance travelled by the light in one year.

1 light year = speed of light × 1 year

= 300000 × 1 year km = 300000 × 365 × 24 × 60 × 60 km

= 9.46×10^{12} km = 9.46×10^{15} m

iii) Parallactic second (or Parsec) : It is the biggest unit of distance.

One parsec is 3.26 times the light year. i.e.,

1 parsec = 3.26 light year = $3.26 \times (9.46 \times 10^{15}$ m) = 3.08×10^{16} m

4. Submultiples of metre

i) decimetre (dm) : One decimetre is one tenth part of a metre

$$1 \text{ dm} = \frac{1}{10} \text{ m} = 10^{-1} \text{ m} = 10 \text{ cm}$$

ii) centimetre (cm) : One centimetre is one hundredth part of a metre.

$$1 \text{ cm} = \frac{1}{100} \text{ m} = 10^{-2} \text{ m} = 1 \text{ cm}$$

Note : 12 inches = 1 foot
3 feet = 1 yard
5 ½ yards = 1 pole or rod
4 pole = 1 chain
80 chain = 1 mile

iii) millimetre (mm) : One millimetre is one thousandth part of a metre.

$$1 \text{ mm} = \frac{1}{1000} \text{ m} = 10^{-3} \text{ m} = 0.1 \text{ cm}$$

iv) micrometre (μm): It is one millionth (10^{-6}) part of a metre.

(micron) $1\ \mu m = 10^{-6}$ m $= 10^{-4}$ cm

v) Nanometre (nm) : It is 10^{-9} th part of a metre.

$$1 \text{ nm} = 10^{-9} \text{ m} = 10^{-7} \text{ cm}$$

vi) Picometre (pm) = 10^{-12} m $= 10^{-10}$ cm

vii) Angstrom (A°) : It is $10^{-10\,th}$ part of a metre.

1 A° = 10^{-10} m = 10^{-8} cm = 10^{-1} nm

$\therefore$ 1 μm = 1000 A° and 1 nm = 10 A°

viii) Fermi (f) or Femto metre : It is $10^{-15\ th}$ part of a metre

It is the smallest unit of distance.

1 fermi = 10^{-15} m

5. **Vernier callipers :** A vernier callipers is also called slide callipers. Vernier callipers was invented by Pierre vernier.

6. Vernier callipers is an instrument by which we can measure the smallest length upto $\frac{1}{10}$th of a mm = 0.1 mm = 0.01 cm.

7. **Principle of vernier :** n V.S.D = (n – 1) M.S.D

Note : Generally 10 V.S.D = 9 M.S.D.

$$1 \text{ V.S.D} = \frac{9}{10} \text{ M.S.D} = \frac{9}{10} \text{ mm [1 M.S.D = 1 mm]} = 0.9 \text{ mm.}$$

Thus, the difference between one main scale division and one vernier scale division is (1 mm – 0.9 mm) = 0.1 mm. Thus we can measure length accurately upto 0.1 mm.

8. **Pitch :** The smallest value of the length or any other unit which can be read directly from a main scale accurately is called pitch.

ASSIGNMENT

1. To measure any physical quantity _____ should be required
   1) Time 2) Mass 3) Unit 4) Length
2. Amount of work done is 10 Joule, here 10 stands for
   1) Unit 2)Numerical value
   3) Both (1) and (2) 4)None of these
3. Which among the following is the international system of units ?
   1) S.I. 2) F.P.S 3) C.G.S 4) M.K.S
4. The mean distance of the earth from the sun is called
   1) Mean solar day 2) Astronomical unit
   3) Light year 4) Parsec
5. Which one of the following is the 10^{-10th} part of a metre?
   1) Micrometre 2) Nanometre 3) Angstrom 4) Fermi
6. Unit of area is a
   1) Derived unit 2) Fundamental unit
   3) Both 1 and 2 4) None of these
7. Derived units are the units of
   1) derived physical quantities 2) fundamental physical quantities
   3) both (1) and (2) 4) None of these
8. One metric tonne = _____ quintal
   1) 10,000 2) 1000 3) 100 4) 10
9. 1 cm^2 = _____
   1) 10^4 m^2 2) $10^3 m^2$ 3) 10^2 m^2 4) 10^2 mm^2
10. How many centimetre make 1 nano metre
    1) 10^{-7} cm 2) 10^{-5} cm 3) 10^{-15} cm 4) 10^{-10} cm
11. 10 kg m^{-3} = _____ g cm^{-3}
    1) 1000 2) 1 3) 10^{-2} 4) 100
12. 16 kg/m^3 = _____ g/cm^3
    1) 16×10^{-3} 2) 26×10^{-6} 3) 36×10^{-3} 4) 46×10^{-6}
13. 1 decade = _____ millennium
    1) $\frac{1}{100}$ 2) $\frac{1}{10}$ 3) 100 4) 1000
14. 0.4 sq. cm is _____ sq. km
    1) 4×10^{-10} 2) 4×10^{-11} 3) 0.4×10^{-11} 4) 4×10^{-9}

15. 1 μg (micro gram) = _____ g
    1) 10^{-10} 2) 10^{-8} 3) 10^{-6} 4) 10^{-4}
16. 10^{14} erg = _____ J
    1) 10^{6} 2) 10^{-7} 3) 10^{7} 4) 10^{2}
17. Vernier callipers consist of two scales they are
    1) metallic scale 2) vernier scale
    3) main scale 4) Both (2) and (3)
18. The magnitude of the smallest measurerment which can be measured by an instruments
    1) least count 2) pitch 3) both (1) &(2) 4) None of these
19. If the zero of the vernier is on the right side of the zero of the main scale then the error is said to be
    1) positive 2) negative 3) zero 4) no error
20. If the zero of the vernier lies between 2.6 cm and 2.7 cm. Then M.S.R is
    1) 2.6 cm 2) 3.6 cm 3) 1 cm 4) None
21. The least count of a vernier callipers is 0.01 cm while measuring the diameter of a cylinder, the M.S.R is 3.60 cm and the 8th V.S.D coincides with the mains scale, then its diameter
    1) 3.68 cm 2) 8.69 cm 3) 13.68 cm 4) 18.67 cm
22. Vernier callipers is also known as
    1) rider callipers 2) slide callipers 3) paul callipers 4) None
23. In a vernier callipers, one main scale is x cm and n divisions of the vernier scale coincide with (n–1) divisions of the main scale. The L.C. (incm) of the callipers is
    1) $\left(\frac{n-1}{n}\right)x$ 2) $\frac{nx}{n-1}$ 3) $\frac{x}{n}$ 4) $\frac{x}{n-1}$
24. 1 km^2 = _____ hectares
    1) 10000 2) 1000 3) 100 4) 10
25. In C.G.S system the unit of area is
    1) s^2 2) kg^2 3) cm^2 4) m^2
26. Unit of area is a
    1) fundamental unit 2) derived unit
    3) both (1) and (2) 4) None
27. The height of a temple tower can be formed by _____ method
    1) Graphical 2) Algebraical 3) Triangulation 4) survey
28. The S.I. unit of volume is
    1) cubic centimetre 2) cubic millimetre
    3) cubic metre 4) cubic litre

LIGHT SHADOWS AND REFLECTION OF LIGHT

1. **Light has an invisible energy:**
2. **Sources of light**:

   Sun is the primary source of light for mankind.In addition to it, lighted bulb, a fluorescent tube,a lighted candle ,a kerosene oil lamp etc.are other sources of light.
3. **Speed of light :** Light travels at very fast speed i.e., 3×10^8 m/s. It means the speed of light is 300000000 m/s or 300000 km/s.
4. **Luminous bodies:-**

   The bodies which give out light energy by themselves are called luminous bodies.

   Ex :- The sun, the stars, glow worm etc.
5. **Non-luminous bodies** :

   The bodies which do not give light energy by themselves , but reflect the light energy falling on them are called non-luminous bodies.

   Ex:- moon,wood,furniture ,etc.
6. **Terms related to light:**

   I) Optical medium: Any material (or) non-material through which light energy passes wholly (or) partially is called optical medium

   Ex:- vaccum,air,water, glass etc.

   II) Homogeneous medium: An optical medium which has a uniform composition throughout is called homogeneous medium.

   Ex:- vaccum,diamond, distilled water, pure alcohol etc.

   III) Heterogeneous medium:- An optical medium, which has different composition at different points is called heterogeneous medium.

   Ex :- Air ,muddy water, fog, mist, clouds, smoke etc .

   IV) Transparent medium:- A medium which allows most of the light energy to pass through it is called transparent medium.

   Ex:- vaccum ,glass,clear air, alcohol, benzene etc.

   V) Translucent medium:- A medium which partially allows the light energy to pass through it is called translucent medium.

   Ex:- oiled paper,tissue paper ,ground glass, butter paper etc.

   VI) Opaque bodies:- The bodies which do not allow the light energy to pass through them are called opaque bodies.

   Ex:-Bricks,wood,metals etc.

   VII) Point source of light :-A source of light which is of the size of pinhead is called point source of light.

   Ex:- The pinhole act as a point source of light.

VIII) Extended source of light :- Any source of light which is bigger than point source of light is called extended source of light

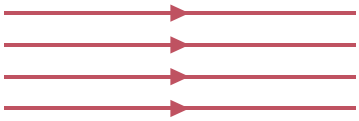
Ex:- Bulb, Tube light, burning candle etc.

IX) Ray of light :- The path along which light energy travels in a given direction is called ray of light .

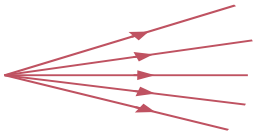
Ray of light

X) Beam of light:- A collection of number of rays of light is called beam of light.

XI) Parallel beam:- When the rays of light travels parallel to each other,then the collection of such rays is called parallel beam.



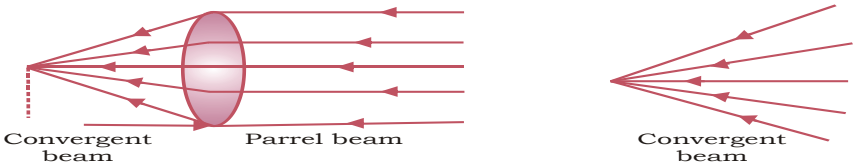
XII) Divergent beam:- When the rays of light originating from a point ,travel in various directions,then the collection of such rays is called divergent beam.



Ex:- The rays coming out from a bulb or a burning candle or a car headlight constitute a divergent beam.

Note:- A point source produces a divergent beam of light.

XIII) Convergent beam:- When the rays of light coming from different directions, meet at a point then the collection of such rays is called convergent beam.



Ex:- If a parallel beam is made to pass through a convex lens, then it meets at a point. This kind of collection of rays is called convergent beam of light.

7. Rectilinear propagation of light:

Light travels in a straight line. This property of light is called rectilinear propagation of light.

8. Shadows and its kinds

a)Shadow

A dark patch behind an opaque body, when it is placed in the path of light is called shadow. A shadow consists of two regions.

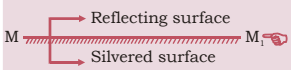
b)Umbra: A region of total darkness is called umbra. No ray of light enters in this region.

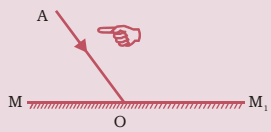
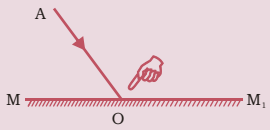
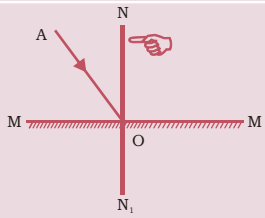
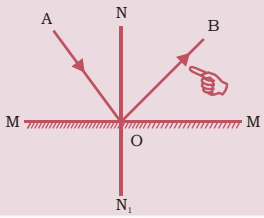
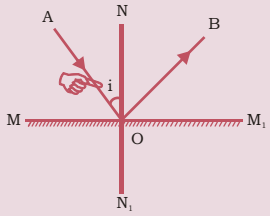
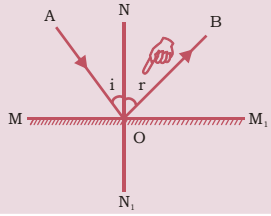
c)Penumbra: A region of partial darkness which surrounds the umbra is called penumbra. Some rays of light always reach this region and partially illuminate

REFLECTION OF LIGHT

1. **Reflection of light:** When a beam of light is incident on a surface, a part of it is returned back into the same medium. The part of light which is returned back into the same medium is called the reflected light.

2. **Terms and Definitions:**

Term	Figure	Representation	Definition
Mirror		MM_1	A highly polished surface which reflects almost whole of the light incident upon it is called a mirror. It has two surface - a reflecting surface and a silver surface

Term	Figure	Representation	Definition
Incident ray		AO	The light ray striking the reflecting surface is called the incident ray
Point of incident		O	The point at which the incident ray strikes the reflecting surface is called the point of incidence
Normal		NN_1	The perpendicular draw to the surface at the point of incidence is called the normal
Reflected ray		OB	The light ray coming back to the same medium after reflection is called reflected ray
Angle of incidence		i	The angle, which the incident ray makes with the normal at the point of incidence, is called the angle of incidence.
Angle of reflection		r	The angle, which the reflected ray makes with the normal at the point of incidence, is called the angle of reflection.

ELECTRICITY AND CIRCUITS

1. **Electric Current**

The rate of flow of charge in a circuit is called electric current. In other words, it is the amount of charge flowing per second. It is denoted by the the letter I.

If Q is the charge which is flowing through a conductor in time t, then current is given by $I = \frac{Q}{t}$

2. **Unit of current:**

The S.I unit of current is ampere and it is denoted by the letter 'A'.

The S.I unit of Q is coulomb and that of t is second.

Thus, the S.I unit of electric current is $\frac{1 \text{ coulomb}}{1 \text{ second}} = 1A$

3. **Definition of ampere:**

When a charge of coulomb flows through a conductor in one second, then the current flowing through the conductor is said to be one ampere.

Thus, when 1 coulomb of charge flows through a conductor in 1 second, then the current flowing through it is said to be 1 ampere.

$1 \text{ ampere} = \frac{1 \text{ coulomb}}{1 \text{ second}}$

4. **Smaller units of electric current:**

Sometimes smaller units of current are also used. These are microampere and milliampere.

1 microampere $= 1\ \mu A = 10^{-6}$ A

1 milliampere = $1\ mA = 10^{-3}$ A

5. **Bigger unit of electric current:**

Sometimes the magnitude of the current flowing in a conductor is very large. This large magnitude of current is expressed in bigger units, such as kilo ampere and mega ampere.

1 kilo ampere (kA) = 1000 A = 10^3 A

1 mega ampere (MA) = 1,000,000 A = 10^6 A.

6. **Flow of current:** In metals, the moving charges are the electrons constituting the current, while in electrolytes and ionized gases, electrons and positively charged ions are the ions moving charges which constitute current.

The charge on an electron is negative and is -1.6×10^{-19} coulomb (symbol C).

Therefore, I C charge is carried by electrons. Hence if I A current flows through a conductor, it implies that 6.25×10^{18} electrons pass in 1 second across the cross section of the conductor.

The direction of current is conventionally taken opposite to the direction of motion of electrons.

If *n* electrons pass through a cross section of a conductor in time t, then total charge passed

Q = n × e and current in conductor

Instrument by which current measured:

Current is measured by an instrument called ammeter.

7. **Potential:** The potential at any point is defined of the work done is bringing a unit positive charge from infinity to that point.

8. **Potential defference :** The potential defference two points in an electric field is defined as the amount of work done in moving a unit positive charge from one point to another point.

$$P.d = \frac{\text{work done}}{\text{Quantity of charge transferred}}$$

If W joules of work has to be done to transfer Q columbs of charge from one point to another then the p.d. 'v between the two point is given by are formula.

$$V = \frac{W}{Q}$$

The S.I unit of p.d. is volt (V)

Primary Cells

1. **Primary Electric cells:**

Voltaic cell, Lechlanche cell, Dry cell and Bichromate cell are few examples of Primary electric cells.

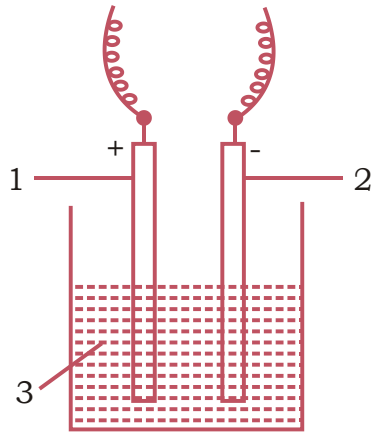
2. **Voltaic cell :**

Construction of a voltaic cell:

a) Take a glass jar and pour dilute sulphuric acid up to 3/4 of the jar.

b) Take a copper plate and a zinc rod and put them in the jar such that half of each of them is immersed in the acid.

c) Take two copper wires and connect each of them to the copper plate and the zinc rod.



Voltaic cell
1. copper sheet(+)
2. zinc sheet(-)
3. dilute sulphuric acid

d) Electricity flows through these wires and if you connect these wires to a electric bell, it rings continuously.

e) In the cell, the copper plate is the positive pole and the zinc rod is the negative pole. The dilute sulphuric acid is called the exciting fluid or electrolyte.

Defects :

a) Local action : Local action due to impure zinc can be prevented by amalgamating the zinc rod with mercury.

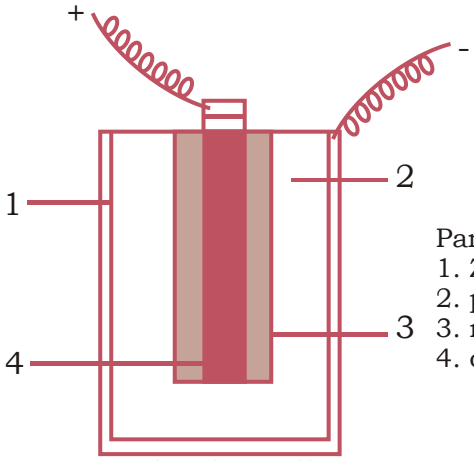
b) Polarisation : Polarisation is the liberation of hydrogen and it can be removed by using oxidising agents or depolarisers like potassium dichromate, copper sulphate or manganese dioxide.

3. Construction of a dry cell:

a) A dry cell consists of a cylindrical zinc can.

b) In the middle of this zinc can, there is a carbon rod with a brass cap.

c) The zinc can acts as the negative pole and the carbon rod acts as the positive pole.



Parts :
1. Zinc con
2. paste of ammontum chloride
3. manganese dioxide + graphite powder
4. carbon rod

The dry cell

ASSIGNMENT

1. The current passing through a conductor is 5 ampere .Then the charge that passes through that conductor in 5 minute is _____Coulomb
   1) 1000 2)1500 3)2000 4)500
2. What is the electric potential at a point in an electric field ,when 96 J work is done in moving a charge of 24C from infinity?
   1) 2V 2)3V 3)5V 4)4V
3. The current through a conductor is 4 ampere . The number of electrons that pass through the conductor in one second is
   1) 3.125×10^{18} 2) 6×10^{18} 3) 25×10^{18} 4) 12.5×10^{18}
4. 25 Coulombs of charge is brought from infinity to a given point in an electric field when 75 joule of work is done .
   1) 3V 2) 2V 3) 4V 4) 5V
5. The primary cell which is used in daily life is
   1) Lechlanche 2) Dry cell 3) Bichrometre cell 4) Simple voltaic cell
6. The depolarization used in leclanche cell is
   1) Solution of ammonium chloride 2) Porous pot
   3) Powered carbon 4) manganese dioxide
7. In a Lechlanche cell the anode is made of
   1) Copper plate 2) Zinc plate 3) carbon plate 4) Iron plate
8. In an voltaic cell the negative pole is
   1) zine rod 2) carbon rod 3) copper rod 4) None of these
9. In an Dry cell the positive pole is
   1) zine rod 2) carbon rod 3) copper rod 4) None of these
10. In an lechlanche cell the positive pole is
    1) zine rod 2) carbon rod 3) copper rod 4) None of these
11. When negative terminal of a cell is connected to the positive terminal of the next cell are said to be in
    1) Series 2) Parallel 3) Both(1) and (2) 4) Neither (1) nor (2)
12. The symbol for electric cell is
    1) 2) 3) 4)
13. The symbol for switch
    1) 2) 3) 4)
14. What is the total emf, when three cells of emfs are 2V,2V,2V are connected in parallel
    1) 4V 2) 1V 3)2V 4) 5V

15. Four electronic cells each of emf 1V connected in parallel , then the combined emf is_____ V

1)1 V 2)2 V 3)3V 4)4 V

16. A device in an electric circuit which can turn on/off the electric circuit is called

1) an electric regulator 2) an electric meter

3) an electric switch 4) all

17. Which is not a small source of ellectric current

1) cells 2) strorage cells 3) button cells 4) electric generator

18. The overhead electric cables used for the trasmission of electric current are made of

1)aluminium 2)zinc 3) steel 4)lead

19. In a simple voltaic cell, the negative plate consists of

1)copper 2)zinc 3) lead 4)tin

20. The major defects of the simple voltaic voltaic cell are called

1)local action 2)polarization 3)neutralisation 4) both (1) and (2)

21. The positive terminal of a simple le-clanche cell consists of

1)zinc 2)carbon 3)copper 4)none

22. The negative terminal of a simple le-clanche cell consists of

1)zinc 2)aluminium 3)iron 4)none

23. The emf of a bichromate cell is about

1)1.75V 2)1.5V 3)1.25V 4)2.0V

24. When a number of cells are connected end to end such that the positive terminal of one cell is connected to negative terminal of the next cell, then the combination of the cells are in

1) series 2)parallel 3)mixed circuit 4)none

25. When the cells are connected in series, the current output of the combination

1) increase 2) decrease 3) remains same 4) none

26. When the number of cells are connected in such a way that they have common positive terminal and common negative terminal, then the cells are connected in

1) series 2)parallel 3)mixed circuit 4)none

27. Copper or aluminium wires donot get hot when electric current flows through them because

1)they are cooled wires 2) They offer no resistance to the electric current

3)They offer resistance to the electric current 4)None

VI CLASS

CHEMISTRY

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CHANGES AROUND US

Change
Changes take place every moment around us.
Examples are the melting of ice, the freezing of water, the digestion of food etc.

Types of changes
1) Reversible and irreversible changes
2) Periodic and non-periodic changes
3) Physical and chemical changes

Reversible change
The change of ice into water and that of water into ice are reversible.
A change is said to be reversible when the change can be reversed by reversing the conditions.

Irreversible change
A change is said to be irreversible when the change **cannot be reversed by** reversing the conditions.

Periodic change
You might have seen a clock with a pendulum. The pendulum moves from side to side, or to and fro, continuously, at fixed intervals of time.
The to-and-fro motion of pendulum represents a periodic change.
Hence, periodic changes are those which will be **repeated at fixed intervals of time.**

Non-periodic change
The melting of ice cream, the curdling of milk, clotting of blood etc., do not repeat themselves at fixed intervals of time. So they are non-periodic changes.
Hence non-periodic changes are those which are **not repeated at fixed intervals of time.**

Physical change

Making of ice cream, Melting of ice, Sublimation of camphor, Balls rolling down Glowing of bulb etc.

For example:
When heated, ice melts to give water. The kind of matter contained in ice and water is the same.
So, no new substances are formed when ice melts. Also, water changes back into ice when cooled.
Thus the change (i.e., the melting of ice) is reversed by reversing the condition (heating is replaced by cooling).
Hence, the melting of ice is a **physical change**.
In all the above examples, no new substances are **formed**.
The changes are **temporary** and easily **reversible**.
A change in which no new substances are formed and which can be reversed

by reversing the conditions is called a **physical change**.

Change of mass in a physical change
No **matter** (mass) is added during a physical change. Similarly, no **matter** (mass) is removed during a physical change.
Therefore, **mass of a substance remains same**.

Chemical change
Burning of petrol, Rusting of iron, Photosynthesis, Formation of wine etc.
For example:
When milk is curdled, a new substance (curd) is formed, and we cannot get back milk from the curd, i.e., the change is irreversible.
Thus, the curdling of milk is a chemical change.

In all the above examples, new substances are **formed**.
The changes are **permanent** and **irreversible**.
A change in which new substances are formed and which cannot be reversed by reversing the conditions is called a **chemical change**.

The mass of the individual substance(s) undergoing a chemical change is altered. For example, if an iron nail is allowed to rust, the mass of the rusted nail is more than that of the original nail.

S.NO	Physical Change	Chemical Change
1	A physical change is temporary	A chemical change is permenanat
2	A physical change is reversible	A chemical change is irreversible
3	No new substance are formed after a physical change	New substance are formed after a chemical change
4	After a physical change the mass of the substance does not change	The mass of the individual substances under goes a chemical change is altered

Physical and chemical changes can take place together. A common example is the burning of a candle. The following changes occur when a candle burns.
The wax under the wick gets heated and melts. The molten wax flows down and solidifies. The changes in state from solid to liquid and again from liquid to solid are **physical changes**.
A part of the molten wax burns out to form new substances- carbon dioxide and water vapour. This change is irreversible. So, the burning of wax is a **chemical change**.
Physical and chemical changes involve energy change
A change in energy takes place when a physical or a chemical change occurs. Energy is **either taken** in (i.e., absorbed) or **given out** (i.e., evolved or emitted) by a substance undergoing a change.
By this we mean that **heat** or **light** is **either absorbed** or **emitted** by a substance during a physical or chemical change.

WATER

Water is very important for all living beings.

Life cannot exist without water.

Most of the body weight of living beings **(nearly 75%)** is made up of water. Water is the most available compound on the earth (three-fourth's of the earth's surface is covered by water).

Suppose for week. Imagine what would happen? Would you be able to cook, clean utensils, wash clothes or bath? What are the other activities you would not be able to do? What would happen if we do not have easy access to water for a long period of time?

Estimation of the amount of water used by a family in a day

Activity	**Amount of water used**
Drinking	
Brushing	
Bathing	
Washing utensils	
Washing clothes	
Toilets	
Cleaning floor	
Any other	
Total water used in a day by a family	

The sources of water

Surface water, i.e., the water in large water bodies like rivers, lakes, oceans and the polar ice caps.

Groundwater, i.e., the water under the surface of the earth.

Water from lakes and rivers (also called fresh water), and groundwater are widely used for agricultural, industrial and household purposes. The source of this water is rain.

Sea water is salty and cannot be used directly.

The States of Water

Water is a **transparent**, **odorless**, **colourless** and **tasteless** liquid. It exists in three states—solid (ice), liquid (water) and gaseous (water vapour). Its presence can be seen in all the three states in the nature.

High mountains and polar regions are covered with ice.

LANGUAGE OF CHEMISTRY

Symbol, Valancy & Formulae

Symbol

A symbol represents short form of an element. The scientist who suggested a method of representing elements using the English letters (Capitals as well as small) is J.J. Berzelius.

A symbol is defined as an abbreviation or short hand sign for the full name of an element.

A symbol represents a specific element.A symbol represents one atom of an element. For example O stands for oxygen, N stands for nitrogen.

A symbol represents the mass of element, equal to its atomic mass of that element.

Quantitatively it represents,

i) One atom of an element.
ii) The number of parts by weight (Atomic weight) of an element.
iii) One gram atom of an element.

For example the symbol O represents

i) One atom of oxygen.
ii) 16 parts by weight of oxygen.
iii) One gram atom of oxygen i.e., 16 grams.

Guidelines to write the symbols

1. For some of the elements the first letter of its English name is used as symbol to represent that element in short form. Only capital letters are used.
2. When the names of the two elements start with the same letter, the second letter or a prominent letter is added to the first letter. When two letters are used the first letter is in capital and the second letter is always a small one.
3. For some elements, the symbols are taken from their latin names.

Atomic Number	Name of the element	Symbol
1	Hydrogen	H
2	Helium	He
3	Lithium	Li
4	Berylium	Be
5	Boran	B
6	Carbon	C
7	Nitrogen	N
8	Oxygen	O
9	Flourine	F
10	Neon	Ne
11	Sodium	Na
12	Magnesium	Mg
13	Aluminium	Al
14	Silicon	Si
15	Phosphorous	P

Atomic Number	Name of the element	Symbol
16	Sulphur	S
17	Chlorine	Cl
18	Argon	Ar
19	Potassium	K
20	Calcium	Ca
21	Scandium	Sc
22	Titanium	Ti
23	Vanedium	V
24	Chromium	Cr
25	Manganese	Mn
26	Iron	Fe
27	Cobalt	Co
28	Nickle	Ni
29	Copper	Cu
30	Zinc	Zn

Symbols for elements :

For some of the elements the first letter of its English name is used as symbol to represent that element in short form. Only capital letters are used.

Example :

S. No.	***Name***	***Symbol***
1.	Hydrogen	H
2.	Carbon	C
3.	Nitrogen	N
4.	Oxygen	O
5.	Fluorine	F
6.	Sulphur	S
7.	Boron	B
8.	Phosphorus	P
9.	Iodine	I

When the names of the two elements start with the same letter, the second letter or a prominent letter is added to the first letter. When two letters are used the first letter is in capital and the second letter is always a small one.

Example :

S. No.	**Element**	**Symbol**
1.	Carbon	C
2.	Calcium	Ca
3.	Cadmium	Cd
4.	Chlorine	C*l*
5.	Phosphorus	P
6.	Platinum	Pt
7.	Boron	B
8.	Barium	Ba

The Symbols of some elements are derived from their Latin names. They are given below.

Example :

S.No.	**Element**	**Latin Name**	**Symbol**
1.	Sodium	Natrium	Na
2.	Potassium	Kalium	K
3.	Iron	Ferrum	Fe
4.	Copper	Cuprum	Cu
5.	Silver	Argentum	Ag
6.	Gold	Auram	Au
7.	Mercury	Hydrargyrum	Hg
8.	Lead	Plumbum	Pb
9.	Tin	Stannum	Sn
10.	Antimony	Stibium	Sb